MALWANDLA HLONGWANE

Junior Database Developer | SQL Developer

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PROFESSIONAL SUMMARY

Final-year Diploma in ICT (Applications Development) student at the University of Mpumalanga with hands-on relational database work across six real applications. Designed a 26-table MySQL schema with 39 foreign keys and 76 indexes, secured a live PostgreSQL database with row-level security and 198 automated database checks, built a tamper-evident audit log, and used transactions and row locks to prevent double bookings. Skilled in ER modelling, normalisation, SQL, indexing, data security and connecting databases to REST APIs. Seeking a Junior Database Developer, SQL Developer, Junior DBA or data-focused graduate role.

TECHNICAL SKILLS

- **Database Systems:** MySQL, PostgreSQL, Supabase, SQLite, Room (Android), MySQL Workbench, pgAdmin, phpMyAdmin
- **SQL:** DDL and DML, complex joins, subqueries, set operators, aggregate and analytical functions, stored procedures, triggers
- **Design:** ER modelling, normalisation, conceptual, logical and physical data models, primary and foreign keys, constraints, indexing
- **Integrity & Performance:** transactions, row-level locking, concurrency control, query optimisation, versioned schema migrations, database testing (pgTAP)
- **Data Security:** row-level security (RLS) policies, prepared statements (SQL injection prevention), password hashing, AES-256 encryption at rest, audit logging, role-based access
- **Application Integration:** PDO, JDBC, JPA/Hibernate, mysql2 (Node.js), REST APIs, CSV export; Java, PHP, JavaScript
- **Also covered in coursework:** distributed databases, data warehousing, star schemas, business intelligence, Big Data and NoSQL (MongoDB, Neo4j, Hadoop)

PROJECT EXPERIENCE

Taskify | Campus Task & Equipment Marketplace

Full-stack web application | MySQL database design
Tech: Node.js, Express, MySQL, JavaScript, HTML, CSS, REST API, JWT, Render

- Designed a 26-table MySQL schema with 39 foreign keys and 76 indexes supporting users, tasks, sales, rentals, disputes and payments.
- Prevented duplicate bookings and payments under concurrent use with row-level locks and conditional updates.
- Combined 28 database repair scripts into one ordered runner and regenerated the schema file, verified against the live database (325 columns) with no rows lost.
- Covered database-driven workflows with 134 automated tests across 13 suites, all passing.

GitHub: github.com/Malwandla21-ict/taskify | **Live site:** taskify-marketplace.co.za

ZAKA | Savings Tracker & Spitori Dictionary

Flutter + Supabase app | PostgreSQL database design | Private repository, code available on request
Tech: Flutter, Dart, Riverpod, go_router, Supabase (PostgreSQL, Auth, Storage, Edge Functions), pgTAP, Cloudflare Pages

- Designed a 17-table PostgreSQL schema in Supabase through 28 versioned migrations, with row-level security policies so each person can only read and change their own data.
- Enforced money rules in the database with constraints and triggers: a goal's balance can never go below R0, daily per-person limits on new records, and a cap on uploaded photos.
- Wrote 198 pgTAP checks across 14 test files, plus a Node.js runner that runs them against the hosted database inside transactions that are always rolled back.

- Kept development and production databases in step by applying the same migrations to both, never changing tables by hand.

Live site: zaka-3hi.pages.dev

StokVault | Stokvel Contribution & Payout Platform
Enterprise Java | Group project (DICT312) | PostgreSQL
Tech: Java 21, Jakarta EE 11 (Faces, REST, EJB, CDI, JPA, JMS, Batch), Payara Server, PostgreSQL, Maven, Docker, GitHub Actions, JUnit 5

- Modelled contributions, payout cycles and per-group member roles in PostgreSQL using JPA entities.
- Built a SHA-256 hash-chained audit log, with a PostgreSQL trigger that blocks UPDATE, DELETE and TRUNCATE on audit records.
- Encrypted ID numbers at rest with AES-256-GCM and enforced uniqueness with an HMAC.

GitHub: github.com/Malwandla21-ict/StokVault

UMP Events | Campus Event Booking Website
PHP web application | Started as a 2nd-year group project, since rebuilt
Tech: PHP 8.2, MySQL, PDO, MVC, HTML, CSS, JavaScript, Apache (XAMPP)

- Moved all SQL into one repository class per table, using PDO prepared statements to prevent SQL injection.
- Prevented overbooking with transactions, row locks and a unique constraint on bookings.
- Built admin reports (fill rates, recent bookings) and CSV attendee exports protected against formula injection.

GitHub: github.com/Malwandla21-ict/Ump-Event-Booking-web

ADDITIONAL PROJECTS

HealthWise | AI Symptom Checker Android App
Native Android application
Tech: Java, Android SDK, XML, Material Components 3, Room (SQLite), Retrofit, Gson, AlarmManager, Hugging Face API

- Designed Room (SQLite) tables for check history, pill reminders, dose logs and health profile, with migrations that keep user data across schema versions.

GitHub: github.com/Malwandla21-ict/HealthWise

FUEL SA | Fuel Ordering & Station Management System
Java desktop application
Tech: Java 17, JavaFX 21, FXML, CSS, SQLite (JDBC), Maven, BulkSMS REST API

- Built an SQLite database with DAO classes over JDBC for customers, vehicles and fuel requests, feeding revenue and top-customer reports.

GitHub: github.com/Malwandla21-ict/FuelSA

EDUCATION

Diploma in Information and Communication Technology (Applications Development)
University of Mpumalanga | Expected completion: 2026 | Academic average: 75–78%
Relevant coursework:

- **Information Systems III (Database Systems):** ER and advanced data modelling, normalisation, SQL programming, join and set operators, database architectures
- **Information Systems III (Advanced Database Concepts):** transaction management, concurrency control, performance tuning, distributed DBMS, database security, Big Data, NoSQL, BI and data warehousing
- **Web Development:** MySQL with PHP (PDO), multi-table queries, foreign keys and joins